Joel Tchouke

Mankato MN tchoukejoel@gmail.com 929-339-7034 Portfolio LinkedIn GitHub

Education

Minnesota State University, Mankato *Jan 2022 – May 2026*
B.S. Computer Engineering — Physics Minor — Honors Program GPA: **3.7/4.0**

- Coursework: Data Structures and Algorithms, Information Security, Computer Architecture, Signals and Systems
- CodePath AI — Applications of AI Engineering
- Dean's List (Fall 2022 – Fall 2024)

Technical Skills

Languages: C++, Python, TypeScript, JavaScript, SQL, Bash
Frameworks: React, Node.js, Express, WebGL, STL
Tools: Docker, Git, Linux, CI/CD, REST APIs, PostgreSQL
Core Areas: Algorithms, Systems Programming, Backend Development, Performance Optimization, Distributed Systems

Experience

Software Engineer Intern — AGCO May 2025 – Dec 2025

- Engineered high-performance **C++ data processing modules**, optimizing memory access patterns and reducing execution latency by **30%** across critical workflows
- Designed and implemented automated validation pipelines, increasing test coverage and reducing regression testing time by **40%**
- Refactored legacy multi-threaded codebases, eliminating race conditions and improving system stability under concurrent workloads
- Diagnosed complex runtime failures using structured logging and debugging tools, reducing issue resolution time by **30%**

Cybersecurity Analyst — Minnesota State University IT Solutions Jan 2025 – Present

- Built Python-based data pipelines to process and analyze large-scale telemetry, reducing manual investigation time by **50%**
- Queried and analyzed logs across **15,000+ endpoints**, identifying anomalies through structured data processing and pattern detection
- Developed reusable automation scripts for log ingestion, parsing, and reporting, improving workflow scalability and reproducibility
- Worked with large datasets and real-time event streams, applying efficient filtering and aggregation techniques

Research Assistant Oct 2023 – Oct 2024

- Developed Python-based simulation and data analysis tools, improving processing efficiency by **25%**
- Implemented algorithmic optimizations for numerical computations, reducing runtime complexity in experimental workflows
- Collaborated in a small team to design scalable software solutions for data-driven research environments

Projects

Full-Stack 3D Portfolio Platform (React, WebGL, Node.js, Docker)

- Built an interactive full-stack platform using **React + WebGL**, rendering real-time 3D environments directly in the browser
- Designed backend APIs with **Node.js/Express** and deployed using containerized infrastructure with **Docker**
- Optimized rendering performance and state management, ensuring smooth user interaction under dynamic workloads
- Structured application architecture for scalability, modularity, and maintainability

High-Performance Systems Application (C++)

- Developed a performance-critical application in **C++**, leveraging efficient memory management and cache-aware data structures
- Optimized algorithms to reduce runtime complexity and improve execution speed on large datasets
- Implemented modular and extensible architecture, improving maintainability and enabling feature expansion

Python Profiling & Performance Analysis Tool

- Built a custom profiling tool to trace execution flow and identify performance bottlenecks in applications
- Collected and visualized runtime metrics, enabling data-driven optimization decisions
- Reduced execution time of tested applications by identifying inefficient code paths and improving resource utilization

Leadership

Cybersecurity Association — President Aug 2024 – Present
IEEE Student Branch — Treasurer May 2024 – Present
MavPASS Leader — Tutor Aug 2023 – May 2025